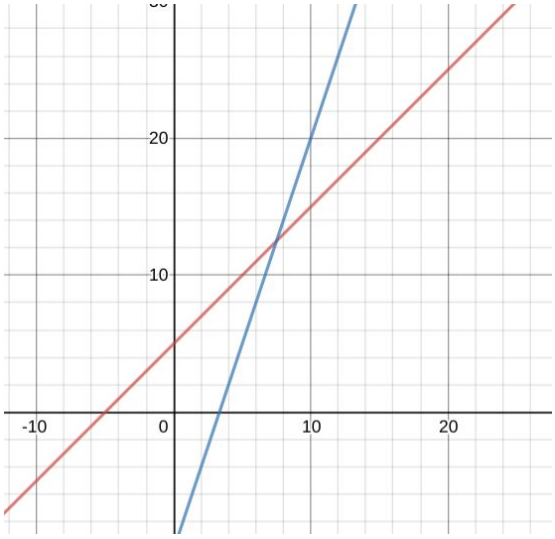
Systems of Equations

<u>Definition of a system of equations</u>:

A **system of equations** is a group of 2+ equations with the same variables. The solution to a system of equations is all of coordinates that satisfy all of the equations. Visually, the solutions are where the equations intersect.



<u>Solving slope-intercept form and slope-intercept form systems: (w/ substitution)</u>

Use the **substitution** technique. Choose one of the equations. This will be the equation you will use to substitute in for y in the other equation. You can do this because the y term is isolated in the equation you chose.

When you substitute in for y, you should only have x terms in the equation now. Then you can use algebra techniques to solve for the x solution.

When you have the x solution, you can plug it into any of the two equations to get the corresponding y value.

You should get the same y value when you plug it into both equations; this is how you know you did it correctly.

- $y = 2x + 3$, $y = x - 5$
  1. $(y) = x - 5$ -> $(2x + 3) = x - 5$ [substitute the y as $2x + 3$]
  2. $2x + 3 = x - 5$
  3. $2x = x - 8$
  4. $x = -8$
  5. $y = 2(x) + 3$ -> $y = 2(-8) + 3 = -16 + 3 = -13$
  6. Solution: (-8, -13)

<u>Solving slope-intercept and standard form systems (w/ substitution):</u>

Use the **substitution** technique. Substitute the slope-intercept equation into the standard form equation's y variable. You can only use the equation in slope-intercept form because its y value has been isolated. (It is possible to use the standard form equation, but it is more difficult, and just a waste of time.)

When you substitute in for y, you should only have x terms in the equation now. Similar to before, when you have the x solution, you can plug it into any of the two equations to get the corresponding y value.

You should get the same y value when you plug it into both equations; this is how you know you did it correctly.

- $y = 2x + 3$, $4x + 3y = 15$
  1. $4x + 3(y) = 15$ -> $4x + 3(2x + 3) = 15$
  2. $4x + 6x + 9 = 15$
  3. $10x + 9 = 15$
  4. $10x = 6$
  5. $x = \frac{3}{5}$
  6. $y = 2(x) + 3$ -> $y = 2(3/5) + 3 = 6/5 + 3 = 21/5$
  7. Solution: $(\frac{3}{5}, \frac{21}{5})$

Solving standard form and standard form systems (w/ substitution):

Use the **substitution** technique. Choose one of the equations and solve for y in terms of x. Now that the equation is in slope-intercept form, you can solve it using the method above.

- $2x + 4y = 10$, $3x - 2y = 11$
  1. $2x + 4y = 10$ -> $4y = 10 - 2x$ -> $y = \frac{10}{4} - \frac{2x}{4}$ -> $y = \frac{5}{2} - \frac{x}{2}$
  2. $3x - 2(y) = 11$ -> $3x - 2(\frac{5}{2} - \frac{x}{2}) = 11$
  3. $3x - 5 + x = 11$
  4. $4x - 5 = 11$
  5. $4x = 16$
  6. $x = 4$
  7. $2(x) + 4y = 10$ -> $2(4) + 4y = 10$ -> $8 + 4y = 10$ -> $4y = 2$ -> $y = ½$
  8. Solution: $(4, \frac{1}{2})$

Solving standard form and standard form systems (w/ elimination):

Use the **elimination** method. *This method is usually faster than **substitution** for solving systems of equations with two standard form equations. This is the preferred method in this scenario*. For this method, choose one of the variables (x or y). It can be any one of them, but usually it is best to choose the most convenient.

Then, find the LCM of the two coefficients next to this variable in the two different equations. Multiply the equations by the value you need to get both of the coefficients equal to the LCM.

Now that the coefficients are equal, you can subtract the two equations by each other. This should cancel out the variable.

Now you should be left with only one variable in the equation. Solve for this variable. Then, substitute in that variable into any of the equations and solve for the other variable.

- $2x + 4y = 10$, $3x - 2y = 11$ (the same equation as above)
    1. $2x + 4y = 10$, $3x - 2y = 11$ -> (*The most convenient variable to choose in this case is y. The LCM in this case is 4. We need to multiply the first equation by 1 to get 4y (which is the same thing) and the second by -2 to get 4y.)*
    2. $1(2x + 4y = 10)$, $-2(3x - 2y = 11)$
    3. $2x + 4y = 10$, $-6x + 4y = -22$
    4. $$\begin{array}{r} (2x + 4y = 10) \\ -(-6x + 4y = -22) \\ \hline 8x + 0 = 32 \end{array}$$
    5. $8x = 32$ -> $x = 4$
    6. $2(x) + 4y = 10$ -> $2(4) + 4y = 10$ -> $8 + 4y = 10$ -> $4y = 2$
    7. $y = \frac{1}{2}$
    8. Solution: $(4, \frac{1}{2})$

Solving any system of equations (w/ graphing):

Use the **graphing** method. This is usually the longest technique and isn't a good use of your time. However, you must be able to understand how to graph linear equations. You must also understand that the solution to a system of equations is where the equations intersect on the graph. This is because the place where the equations intersect fulfills both of the equations.

Systems of Inequalities:

Definition of a system of inequalities:

Systems of inequalities are similar to systems of equations. The solution to a system of inequalities is the group of coordinates that satisfy both of the inequalities. Visually, this is represented by the region where they intersect. Since inequalities are not lines, but areas, this point of intersection is a region, not just a single point.

Solving systems of inequalities (w/ graphing):

Use the **graphing** method. This is the primary method used to solve systems of equations. You can't use the substitution or elimination methods because you are dealing with inequalities instead of equations, which can't be simplified like that.

Inequalities, when graphed, represent areas instead of just lines. The region where the areas intersect is the group of solutions to your system.

You can graph the boundary to linear inequalities by treating them as linear equations. First, plot the inequality as if it was a linear equation. If the equation is in standard form, then you will want to convert it from standard form into slope-intercept form.

If the inequality has a $\leq$ *or* $\geq$, draw it as a solid line. This means that the points on the line are included within your inequality. If the inequality only has a $<$ *or* $>$, draw it as a dotted line. This means that the points on the line are NOT included within your inequality.

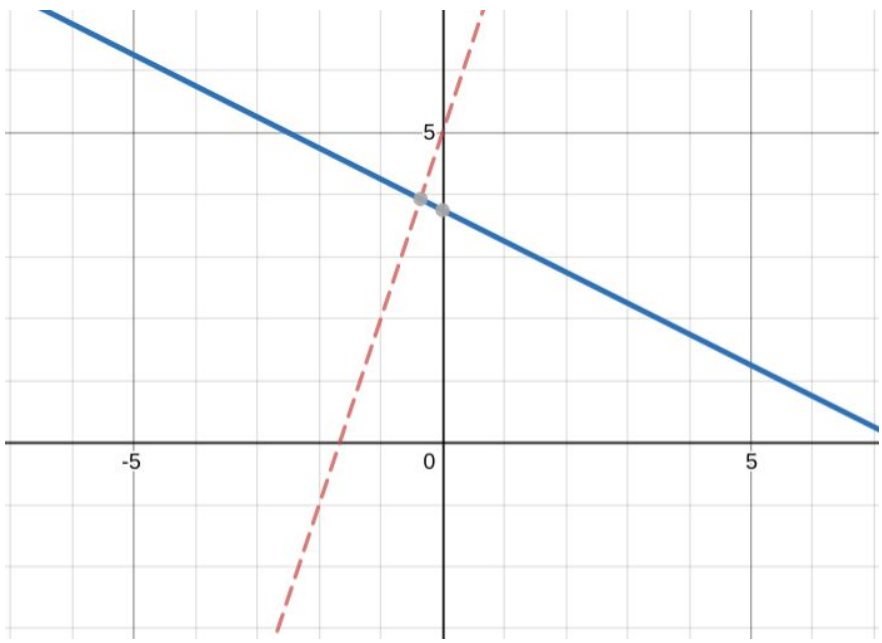
We aren't done, though! We've only graphed the boundary of the inequality! We have to graph the whole region.

When doing this, make sure to graph different equations with different patterns for their regions. For example, for one equation, you might graph the region with horizontal lines and graph the other with vertical lines. This makes it clear where the two inequalities intersect.

Now that we know the boundary of our inequality, we have to determine if the region of the inequality is above or below that boundary. If the inequality has $y >$... or $y \geq \ldots$, then the region of the inequality is above the boundary. If it has $y <$ ... or $y \leq \ldots$, then the region of the inequality is above the boundary.

If it is a vertical line, represented by an inequality like $x > 5$, repeat these steps, but "above" is simply to the right since this is where the x values are bigger, and "below" is simply to the left since this is where the x values are smaller.

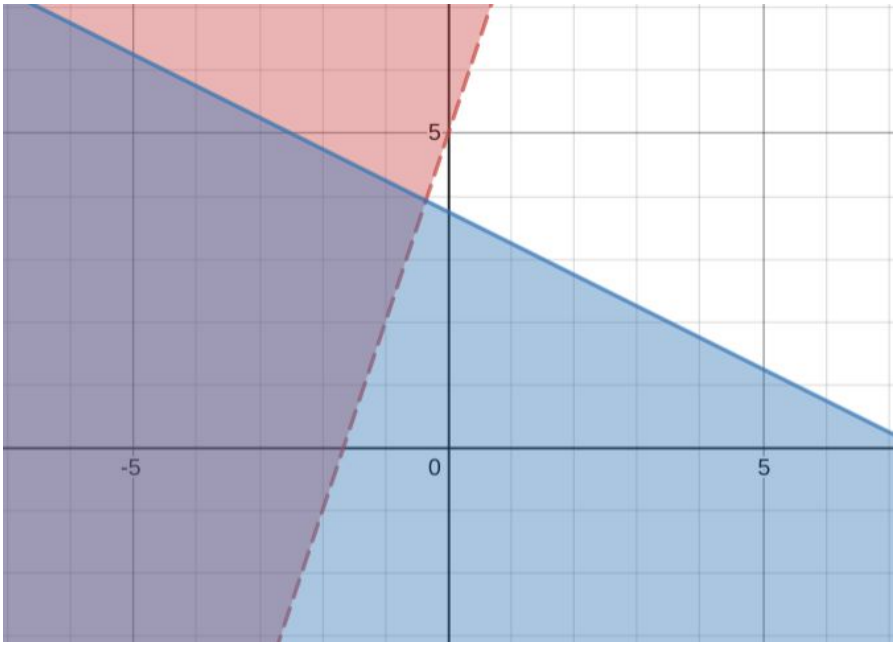
- $y > 3x + 5, 2x + 4y \leq 15$
- $y > 3x + 5, 2x + 4y \leq 15 \rightarrow 4y \leq 15 - 2x \rightarrow y \leq \frac{15}{4} - \frac{x}{2}$ (*This is in a form we can use now.*)
- $y > 3x + 5, y \leq \frac{15}{4} - \frac{x}{2}$
- $y = 3x + 5, y = \frac{15}{4} - \frac{x}{2}$ (*Now we graph the boundaries.*)
- (*Now we can graph our lines. Make sure to graph the first line as a dotted line, since it has a > symbol, which means it doesn't include places where* $y = 3x + 5$. *Make sure to graph the second line as a solid line, since it has a* $\leq$ *symbol, which means it does include places where* $y = \frac{15}{4} - \frac{x}{2}$.*)*



1 $y = 3x + 5$

2 $y = \frac{15}{4} - \frac{x}{2}$

- (*Now we graph the regions. The first inequality's region is above the line because it has a > symbol. The second inequality's region is below the line because it has a* $\leq$ *symbol.*)



1 $y > 3x + 5$

2 $y \leq \frac{15}{4} - \frac{x}{2}$

- The purple region is where the two inequalities intersect. This is the solution to the system of inequalities.

Word Problems with Systems of Equations

Word problems are a common type of systems of equations question. To solve these problems, you will have to be able to represent the situation with equations. Here is an example question:

Example 1:

Allie has a brother named John. Allie is 1 year younger than someone twice John's age. Allie is also seven years older than John. How old are Allie and John?

In this problem, we can represent Allie's age with the variable a, and John's age with the variable j. The first piece of information we are given tells us the following:

$$a = 2j - 1$$

The second piece of information gives us another equation.

$$a = j + 7$$

Since these are both linear equations, we can solve the system to find the values of a and j that fulfill both of these equations (since both of them must be true due to the information the problem gave us). These are both in slope-intercept form, so we can use the substitution method.

$$2j - 1 = j + 7$$
$$j - 1 = 7$$
$$j = 8$$
$$a = j + 7 = 15$$

This means that John is 8 years old and Allie is 15 years old.
To check our work, we can plug it back into our original equations. Also, we can look at the problem to see if it makes sense, since these equations are just representing what our problem is saying. Is Allie 1 year younger than twice John's age? Yes. Is Allie 7 years older than John? Yes, so our answer is correct.

Example 2:
The school held a play yesterday. The play, Romeo and Juliet, sold 300 tickets in total and raised $600. All proceeds went to the local can drive. Admission for children cost $2.50 and admission for adults cost $3. How many children's tickets were sold?

This time, it's a bit harder to figure out what the equations are. First, let's figure out our variables. The thing we want to find is the number of children's tickets sold, and it's an unknown amount, so that'll be our first variable,

c. We can see there's another variable amount, which is the number of adult tickets, which is our second variable, *a*.

Now we have to figure out our equations. There are three other important pieces of information given to us. First, we know the play sold 300 tickets total. We also know it raised $600. Also, we know children's tickets cost $2.50 and adult tickets cost $3.

The first piece of information tells us there were 300 tickets sold in total. This means that that is the amount of children's tickets and adult tickets combined, so we get the following equation:

$$a + c = 300$$

We can do a similar thing with the second piece of information. However, it's more difficult to figure out how to represent the money raised in terms of *a* and *c*. We have to use the third piece of information. Since each children's ticket costs $2.50, the total amount raised by the children's tickets is *2.50*c*, or *2.5c*. Since each adult ticket costs $3, the total amount raised by the adult tickets is *3*a*, or *3a*. We get the following equation:

$$3a + 2.5c = 600$$

These are both in standard form, so we can solve this equation using elimination.

$$a + c = 300$$
$$3a + 2.5c = 600$$

The easiest variable to use is *a,* since the LCM of the two coefficients is just 3. One of the equations already has *3a*, so we just have to multiply the other by 3.

$$\begin{array}{rl} & 3a + 3c = 900 \\ - & (3a + 2.5c = 600) \\ \hline & 0.5c = 300 \end{array}$$

$$0.5c = 300$$

$c = 600$

Now we get our answer: 600 children's tickets were sold.

Word Problems with Systems of Inequalities:

These are also known as constraint problems.

Example 1:
The company Big Red Apple sells both pencils and pens. Pencils are sold for $2 a pack, while pens are sold for $3 a pack. They only have 50 packs in total to sell. It cost them $100 to produce all of them in total. Set up a system of inequalities and show the possible sales they can make that will allow them to break even or make a profit.

First, let's figure out our variables. Since this is a system of inequalities, we'll eventually have to graph it. So when we figure out our variables, one will have to be x and one will have to be y. Let's let the number of packs of pencils sold be x and the number of packs of pens sold be y.

The first inequality we can make is based on the information that there are only 50 packs in total to sell. This means the company can't sell more than 50 packs, so we get the following inequality:

$x + y \leq 50$

The second inequality we can make is based off of the information that the company's sales have to break even or make a profit. This means the company has to sell enough pencils or pens to make an amount equal to or more than the cost of production, which was $100. This gives us the following inequality:

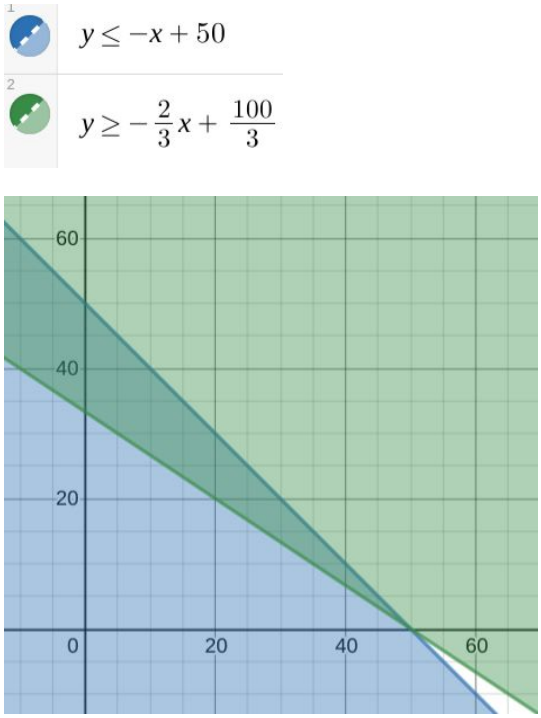
$2x + 3y \geq 100$

Now, to graph our system of inequalities, we have to put both of the inequalities into slope-intercept form.

$y \leq -x + 50$

$y \geq -\frac{2}{3}x + \frac{100}{3}$

And we get the following constraint when we graph it:



Ignore the negative quadrants of the graph, since we can't have a negative amount of pencils or pens. The dark green area is our answer. This means that selling 5 packs of pencils and 40 packs of pens, for example, is a valid sale the company can make that fulfills the conditions of the problem.

(*An additional note: the fact that the number of pencils and pens sold can't be negative acts as another constraint, which is* $x \geq 0$ *and* $y \geq 0$. *When we exclude the negative quadrants, we're just including those additional conditions).*